

A

In-silico PCR per assay

AA1 assay	90 bp	3	≥4	≥4	3	3	≥4
AA2 assay	≥4	145 bp	≥4	≥4	3	3	≥4
AA3 assay	≥4	3	145 bp	3	3	3	3
AA4 assay	≥4	≥4	≥4	125 bp	3	≥4	3
AA5 assay	≥4	≥4	≥4	≥4	107 bp	≥4	≥4
AA6 assay	≥4	3	≥4	3	3	112 bp	≥4
AA7 assay	≥4	3	≥4	≥4	≥4	≥4	95 bp
	AA1	AA2	AA3	AA4	AA5	AA6	AA7

genome (all replicons)

B

Knock-out test (pooled primers)

AA1 masked	lost						
AA2 masked		lost					
AA3 masked			lost				
AA4 masked				lost			
AA5 masked					lost		
AA6 masked						lost	
AA7 masked							lost
	AA1	AA2	AA3	AA4	AA5	AA6	AA7

assay

A: blue = in-silico PCR product (designed length); grey cells give the fewest mismatches of any forward/reverse primer site in that genome (bowtie1 exhaustive search, ≥ 4 = no site with ≤ 3). No off-target product with ≤ 3 mismatches per primer.

B: each row masks one target amplicon; blue = product still formed; only the masked target is lost.